

Observability Limits in Cyber Defense: Distinguishing Distributed Swarm Offensives from Monolithic Attack Streams Through High-Fidelity Telemetry

Abstract

Contemporary cyber operations increasingly rely on distributed infrastructures composed of botnets, compromised cloud assets, proxy networks, and decentralized operator collectives. While these architectures provide resilience and scale for adversaries, they also introduce a critical challenge for defensive cyber operations (DCO): determining whether observed malicious activity originates from a coordinated swarm of heterogeneous actors or from a single dominant source.

This paper argues that the problem is fundamentally one of observability rather than simple traffic volume analysis. Under conditions of limited telemetry resolution, insufficient feature extraction, or excessive aggregation, distinct attack sources become indistinguishable within the defender's observation space. The resulting loss of separability can lead to attribution errors, inefficient defensive resource allocation, and flawed operational responses.

A theoretical framework is presented that models attack attribution as an inference problem constrained by telemetry fidelity. The analysis demonstrates how improvements in temporal resolution, feature diversity, and distributed sensing increase the defender's ability to distinguish coordinated swarm activity from monolithic attack streams. The paper concludes by proposing an observability-centric approach to cyber defense that prioritizes high-fidelity telemetry and feature-rich analytics over aggregate traffic measurements.

1. Introduction

Cyber conflict increasingly occurs within environments characterized by massive scale, high velocity, and substantial uncertainty. Modern adversaries rarely rely upon singular attack infrastructures. Instead, they employ distributed architectures that include botnets, cloud-hosted transient assets, compromised edge devices, anonymization services, and geographically dispersed command-and-control mechanisms.

Despite this evolution, many defensive systems continue to evaluate hostile activity through aggregated measurements that emphasize traffic volume, alert counts, or summarized event streams. Such approaches create a significant analytical risk. Multiple independent attack sources may appear indistinguishable from a single coordinated source when observed through telemetry systems operating with insufficient temporal or informational resolution.

The operational consequences are substantial. Defensive resources may be concentrated against a perceived central attacker that does not actually exist. Attribution assessments may incorrectly identify the nature or structure of an adversary. Mitigation efforts optimized for monolithic attacks may prove ineffective against distributed campaigns.

This paper proposes that these failures should be understood as limitations of observability. Specifically, the defender's ability to distinguish attack structures depends upon the quality, frequency, and diversity of measurements available within the observation space.

2. Cyber Attack Attribution as an Observability Problem

In systems theory, observability refers to the extent to which the internal state of a system can be inferred from external measurements.

Applied to cyber defense, the true adversarial environment exists as a hidden state:

$$X = \{x_1, x_2, \dots, x_n\}$$

where each x_i represents an individual attacking entity possessing unique characteristics such as:

- Infrastructure origin
- Network behavior
- Temporal activity patterns
- Protocol usage
- Payload construction
- Operational objectives

Defenders do not directly observe X . Instead, they observe telemetry:

$$Y = H(X) + \varepsilon$$

where:

- H represents the measurement process
- ε represents noise, uncertainty, loss, and adversarial deception

The central challenge becomes determining whether the observed telemetry Y contains sufficient information to infer the underlying structure of X .

If multiple adversarial configurations produce nearly identical observations, attribution confidence necessarily declines.

3. Feature-Space Collapse and Attack Indistinguishability

A distributed swarm attack can be conceptualized as a collection of attack vectors occupying a multidimensional feature space.

Each attack source contributes a feature vector:

$$f_i = [t, p, h, r, b, \dots]$$

where dimensions may include:

- Timing characteristics
- Packet metadata
- Header fingerprints
- Routing properties
- Behavioral indicators

The defender attempts to classify these vectors into distinct clusters representing unique attack entities.

However, telemetry systems frequently perform aggregation operations that reduce dimensionality:

$$G(f_i) \rightarrow z$$

Examples include:

- Flow summarization
- Alert correlation
- Log batching
- Traffic aggregation

As dimensionality decreases, previously separable clusters may overlap.

When this occurs, multiple attack populations become observationally indistinguishable.

The defender experiences what can be described as feature-space collapse: the loss of sufficient information required to separate distinct adversarial actors.

The practical result is the perception of a single attack stream where multiple coordinated entities actually exist.

4. Telemetry Fidelity and Attribution Confidence

The ability to distinguish swarm activity depends on three interacting factors.

4.1 Temporal Resolution

Attack differentiation requires observations at a temporal granularity sufficient to preserve behavioral variation.

High-frequency telemetry captures:

- Burst structures
- Synchronization patterns
- Propagation timing
- Command-and-control coordination

Low-frequency telemetry obscures these signals through averaging.

As observation intervals increase, multiple independent attack sequences converge toward similar statistical representations.

4.2 Feature Diversity

Traffic volume alone provides limited attribution value.

Effective differentiation requires feature extraction across multiple dimensions including:

- Network-layer behavior
- Transport-layer characteristics
- Application-layer signatures
- Infrastructure relationships
- Historical activity patterns

The probability of distinguishing attack sources increases as independent feature dimensions are added.

4.3 Sensor Distribution

Centralized monitoring introduces blind spots.

Distributed sensing architectures improve observability by capturing adversarial behavior from multiple perspectives simultaneously.

Differences in:

- Route paths
- Latency profiles
- Geographic visibility
- Autonomous system traversal

can reveal distinctions that remain hidden within centralized datasets.

5. Attribution Confidence Under Telemetry Constraints

Attribution should not be treated as a binary outcome but as a confidence function:

$$C = F(T, D, S)$$

where:

- T represents temporal resolution
- D represents feature diversity
- S represents sensor coverage

Confidence increases as telemetry quality improves.

Conversely, attribution uncertainty grows when observations become sparse, aggregated, or noisy.

Importantly, distributed attacks do not inherently prevent attribution. Rather, attribution difficulty emerges when telemetry quality falls below the threshold required to preserve attack separability.

This distinction reframes the problem from one of adversary complexity to one of measurement capability.

6. Operational Consequences

The inability to distinguish attack structures directly impacts defensive operations.

Misallocation of Defensive Resources

Defenders may concentrate inspection capacity against a perceived dominant source while distributed attackers continue operating through unaffected channels.

Faulty Mitigation Strategies

Techniques effective against monolithic attacks may fail against distributed campaigns.

Examples include:

- Static blocking
- Route suppression
- Single-node disruption

Distributed offensives often require:

- Behavioral filtering
- Distributed mitigation
- Edge-based enforcement
- Adaptive rate control

Attribution Error

Incomplete observability increases the risk of associating activity with incorrect actors, infrastructures, or sponsoring organizations.

Such errors can produce strategic consequences disproportionate to the original attack.

7. Toward an Observability-Centric Defensive Architecture

Future defensive cyber operations should prioritize observability as a primary design objective.

Key principles include:

1. High-frequency telemetry collection.
2. Multi-layer feature extraction.
3. Distributed sensing architectures.
4. Real-time correlation across observation domains.
5. Probabilistic attribution models incorporating uncertainty.

Rather than relying primarily on aggregate indicators, defenders should maximize preservation of information throughout the telemetry pipeline.

The objective is not simply detecting attacks but maintaining sufficient observational fidelity to infer their underlying structure.

8. Conclusion

The distinction between a distributed swarm offensive and a monolithic attack stream is fundamentally an observability problem. Attribution failures frequently arise not because adversaries are inherently unknowable but because telemetry systems fail to preserve the information required to differentiate them.

As cyber operations continue to increase in scale and complexity, defensive success will depend upon maintaining high-fidelity observations across time, protocol layers, and geographic perspectives. Organizations capable of preserving feature separability within their telemetry pipelines will possess a decisive advantage in identifying, attributing, and countering distributed adversarial campaigns.

The future of cyber defense is therefore not merely a question of stronger mitigation technologies but of improved observability. Accurate understanding of adversarial structure begins with the ability to measure it.